Application E3 - Problem Statement:

A projectile of mass $m = 100$ g is thrown from a point O with a speed $V_0 = 10$ m/s at an angle $\alpha = 30°$ *with the horizontal*, as shown. Point O is at a height $h = 1$ m above the ground. Friction is negligible. The GPE=0 reference is the ground. The y-axis is positive *downwards*. (We will use $g = 10$ m/s$^2$).

**a.** Determine the speed of the projectile at point D just before reaching the ground. **b.** Determine the $P_x$ and $P_y$ components of the momentum vector at time $t$. **c.** Deduce the expressions of $V_x$ and $V_y$ at the instant $t$. **d.** Determine the time equations of motion $x(t)$ and $y(t)$. **e.** At what moment does the projectile reach the ground in D?

Initial Setup & Units:

- Mass ($m$): $100 \text{ g} = 0.1 \text{ kg}$
- Initial Velocity ($\vec{V_0}$): The vector $\vec{V_0}$ points down and right.
  - $V_{0x} = V_0 \cos(\alpha) = 10 \cdot \cos(30°) \approx 8.66$ m/s
  - $V_{0y} = V_0 \sin(\alpha) = 10 \cdot \sin(30°) = 5$ m/s (positive, as it's in the +y direction)
  - $\vec{V_0} = (8.66\vec{i} + 5\vec{j})$ m/s
- Initial Position: At $t = 0$, the projectile is at the origin O, so $x(0) = 0$ and $y(0) = 0$.
- Position of Ground (D): The ground is $h = 1\text{m}$ *below* the origin O. In our coordinate system, this is at $y = 1$ m.

a. Speed at Ground ($V_D$)

Since friction is negligible, mechanical energy is conserved. We'll compare the energy at O (initial) and D (final).

- Energy at O ($ME_O$):
  - $KE_O = \frac{1}{2}mV_0^2 = \frac{1}{2}(0.1)(10)^2 = 5$ J
  - $GPE_O = mgh_O$. Point O is $1$ m *above* the GPE=0 level.
  - $GPE_O = (0.1)(10)(1) = 1$ J
  - $ME_O = KE_O + GPE_O = 5 + 1 = 6$ J
- Energy at D ($ME_D$):
  - $KE_D = \frac{1}{2}mV_D^2 = \frac{1}{2}(0.1)V_D^2 = 0.05V_D^2$
  - $GPE_D = 0$ (at the reference level).
  - $ME_D = 0.05V_D^2$
- Conservation of Mechanical Energy ($ME_O = ME_D$):
  - $6 = 0.05V_D^2$
  - $V_D^2 = \frac{6}{0.05} = 120$
  - $V_D = \sqrt{120} \approx 10.95$ m/s

b. Momentum Components ($P_x, P_y$) at time $t$

We use the **Theorem of Linear Momentum**: $\Delta\vec{P} = \Sigma\vec{F}_{ext} \cdot \Delta t$. The only external force is gravity: $\Sigma\vec{F}_{ext} = m\vec{g}$. Since the +y axis is downwards, $\vec{g} = g\vec{j} = 10\vec{j}$.

- $\vec{P}_t - \vec{P}_0 = (m\vec{g}) \cdot t$
- $\vec{P}_t = \vec{P}_0 + (m\vec{g})t$

First, find the initial momentum vector $\vec{P}_0$:

- $\vec{P}_0 = m\vec{V}_0 = 0.1 \cdot (8.66\vec{i} + 5\vec{j}) = 0.866\vec{i} + 0.5\vec{j}$

Now, substitute into the theorem:

- $\vec{P}_t = (P_x\vec{i} + P_y\vec{j}) = (0.866\vec{i} + 0.5\vec{j}) + (0.1 \cdot 10\vec{j})t$
- $\vec{P}_t = (0.866\vec{i} + 0.5\vec{j}) + (1\vec{j})t$
- $\vec{P}_t = 0.866\vec{i} + (0.5 + t)\vec{j}$

By comparing components, we get:

- $P_x = 0.866 \text{ kg} \cdot \text{m/s}$ (This is constant, as there is no horizontal force)
- $P_y = 0.5 + t \text{ kg} \cdot \text{m/s}$

c. Velocity Components ($V_x, V_y$) at time $t$

We can find the velocity components directly from the momentum components, since $\vec{P} = m\vec{V} \implies \vec{V} = \frac{\vec{P}}{m}$.

- $V_x = \frac{P_x}{m} = \frac{0.866}{0.1} = 8.66 \text{ m/s}$ (Constant, as expected)
- $V_y = \frac{P_y}{m} = \frac{0.5+t}{0.1} = 5 + 10t \text{ m/s}$

d. Equations of Motion ($x(t), y(t)$)

We find the position equations by taking the anti-derivative (integral) of the velocity components.

- x-position:
  - $V_x = \frac{dx}{dt} = 8.66$
  - $x(t) = \int 8.66 \, dt = 8.66t + C_1$
  - At $t = 0$, $x = 0$ (at origin O). So $C_1 = 0$.
  - $x(t) = 8.66t$
- y-position:
  - $V_y = \frac{dy}{dt} = 5 + 10t$
  - $y(t) = \int (5 + 10t) \, dt = 5t + 10\left(\frac{t^2}{2}\right) + C_2 = 5t + 5t^2 + C_2$
  - At $t = 0$, $y = 0$ (at origin O). So $C_2 = 0$.
  - $y(t) = 5t^2 + 5t$

e. Time to Reach Ground ($t_D$)

The projectile reaches the ground at point D. From our setup, we know the ground is at the y-coordinate $y = 1 \text{ m}$. We just need to find the time $t_D$ when $y(t_D) = 1$.

- $y(t_D) = 5t_D^2 + 5t_D$

- $1 = 5t_D^2 + 5t_D$
- $5t_D^2 + 5t_D - 1 = 0$

This is a quadratic equation. We use the formula $t = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$:

- $t_D = \frac{-5 \pm \sqrt{5^2 - 4(5)(-1)}}{2(5)}$
- $t_D = \frac{-5 \pm \sqrt{25 + 20}}{10}$
- $t_D = \frac{-5 \pm \sqrt{45}}{10}$

Since time must be positive, we take the positive root:

- $t_D = \frac{-5 + \sqrt{45}}{10} \approx \frac{-5 + 6.708}{10}$
- $t_D \approx \frac{1.708}{10}$
- $t_D \approx 0.171 \text{ s}$

*(Note: Your PDF's solution for this step (t=0.162s) has a small numerical error in its $V_y$ calculation. This method, using the position equation, is more direct and less prone to rounding errors.)*